

AI-Assisted Financial Risk Profiling and Personalized Investment Decision Support

Author Name 1, Author Name 2, Author Name 3

Department of _

Institution Name, City, Country

Email: _

Abstract

Making appropriate investment decisions requires individuals to consider their financial capacity, investment objectives, investment horizon, and tolerance for financial risk. However, individuals often have access to financial information without having an effective mechanism for converting that information into understandable and personalized investment insights. This paper presents an AI-assisted financial decision-support system that combines financial analytics, risk profiling, investment-category matching, and an AI agent. The proposed system collects basic financial information including income, expenses, savings, investment capacity, financial goals, investment horizon, and risk-preference responses. These inputs are processed to derive financial indicators and determine the user's risk profile. A rule-based recommendation mechanism then maps the resulting profile to suitable investment categories. An AI agent equipped with specialized financial-analysis tools orchestrates the analytical process and provides natural-language explanations of the resulting insights. The system is implemented using a web-based architecture with a React frontend, Python-based backend, PostgreSQL database, and large language model API. The proposed approach is evaluated using synthetic financial profiles and user-oriented evaluation criteria including risk-profile consistency, recommendation relevance, explanation clarity, and usability. The study demonstrates a practical approach for transforming structured financial data into personalized and interpretable decision support while maintaining a distinction between AI-assisted guidance and autonomous financial decision-making.

Keywords: Financial Analytics, Artificial Intelligence, FinTech, Risk Profiling, Investment Decision Support, Personalized Finance

1. INTRODUCTION

The increasing availability of digital financial information has created opportunities for individuals to use data-driven approaches when making financial decisions. Income, expenditure, savings, investment capacity, financial goals, and investment preferences provide useful information for understanding an individual's financial situation. However, the availability of such information does not necessarily mean that users can interpret it effectively or determine how it should influence their investment decisions.

Financial analytics can help convert raw financial information into meaningful indicators. Measures such as savings rate, expense-to-income ratio, disposable income, investment capacity, and financial goal

requirements can provide a structured representation of an individual's financial position. Risk profiling can further identify how an individual responds to investment uncertainty and potential losses.

Artificial intelligence provides an additional opportunity by allowing analytical results to be communicated in natural and understandable language. Instead of requiring users to interpret numerical outputs independently, an AI-based decision-support component can explain the factors influencing a recommendation and allow users to interact with the analytical results.

The proposed project therefore focuses on a specific application of artificial intelligence in financial analytics: **AI-assisted financial risk profiling and personalized investment decision support**. The system does not attempt to predict stock prices, execute financial transactions, or autonomously manage investments. Instead, it combines structured financial analytics and a controlled AI agent to assist users in understanding their financial profile and identifying investment categories that may be aligned with their stated characteristics.

1.1 Problem Statement

Individuals may make investment decisions without systematically considering their financial capacity, investment horizon, financial objectives, and tolerance for risk. Conventional financial applications may provide calculators or investment information, but users may still require assistance in interpreting these outputs.

Therefore, the problem addressed by this project is:

How can personal financial information and risk preferences be analysed and combined with an AI agent to provide personalized and interpretable investment decision support?

1.2 Objectives

The main objective of this study is to develop an AI-assisted financial decision-support system that analyses personal financial information, determines an individual's risk profile, and provides personalized investment-category guidance.

The specific objectives are:

1. To collect and process basic personal financial information.
2. To derive meaningful financial indicators from the collected data.
3. To assess the user's investment risk profile using a structured questionnaire.
4. To match the resulting profile with suitable investment categories.
5. To develop an AI agent capable of using predefined financial-analysis tools.
6. To provide understandable explanations for analytical outcomes.
7. To evaluate the consistency, relevance, and usability of the proposed system.

1.3 Scope

The project focuses on financial profile analysis, risk assessment, investment-category matching, and AI-assisted explanation. It does not include automated investment execution, stock-price prediction, guaranteed-return prediction, tax optimization, legal document generation, or autonomous portfolio management.

2. RELATED WORK AND RESEARCH GAP

Financial analytics has increasingly been applied to support financial planning, investment analysis, and decision-making. Personal financial applications commonly use income, expenditure, savings, and investment information to provide users with financial summaries and analytical indicators.

Risk profiling is another important component of investment decision-making. An individual's willingness to accept investment uncertainty can differ substantially based on financial circumstances, investment horizon, objectives, and personal preferences. Consequently, investment decision-support systems need to consider more than historical investment performance alone.

Artificial intelligence has expanded the capabilities of financial applications by enabling natural-language interaction, recommendation systems, and intelligent decision support. More recent AI systems can combine language models with external tools, allowing the AI component to request structured calculations rather than relying exclusively on generated text.

However, a practical challenge remains in connecting these components into a controlled workflow. A financial decision-support system should ensure that numerical calculations and risk classifications are generated by deterministic analytical mechanisms, while the AI component is primarily responsible for orchestrating available tools and explaining the results.

Table 1. Comparison of Conventional and Proposed Approaches

Capability	Basic Financial Application	AI Chatbot	Proposed Approach
Financial data collection	✓	Limited	✓
Financial calculations	✓	Limited	✓
Risk profiling	Limited	Variable	✓
Investment-category matching	Limited	Variable	✓
Tool-based AI processing	–	Limited	✓
Personalized explanation	Limited	✓	✓
Controlled recommendation logic	✓	Limited	✓

The research gap addressed by this project is therefore the development of a **focused AI-agent-based workflow that connects structured financial analytics and risk profiling with personalized investment decision support while maintaining controlled recommendation logic.**

3. PROPOSED METHODOLOGY

The proposed methodology follows a sequential process in which user financial information is converted into analytical indicators, a risk profile is generated, investment categories are identified, and an AI agent provides an interpretable explanation.

Figure 1. Proposed Financial Decision-Support Framework

[INSERT FIGURE 1 HERE]

Figure placeholder:
Financial Data → Data Processing → Financial Analytics → Risk Assessment → Investment Matching → AI Agent → Personalized Decision Support

The methodology consists of the following stages.

3.1 Financial Data Collection

The system collects a limited set of financial and personal variables required for analysis.

Table 2. Input Variables

Category	Variables
Personal	Age
Income	Monthly income
Expenses	Monthly expenditure
Savings	Current savings
Investment	Planned investment amount
Goal	Primary financial objective
Horizon	Expected investment duration
Risk	Risk-preference questionnaire responses

Only information required for the analysis is collected. Sensitive information that is not necessary for the research objective is excluded.

3.2 Data Processing

The collected data undergoes basic validation and preprocessing before being passed to the analytical components.

The processing stage includes:

- checking for missing values;
- validating numerical values;
- identifying inconsistent inputs;
- standardizing financial values;
- calculating derived financial indicators;
- converting questionnaire responses into numerical scores.

The processed information is then stored in structured form for subsequent analysis.

3.3 Financial Analytics

The financial analytics module converts the user's raw financial information into interpretable indicators.

Savings

The basic monthly savings amount is calculated as:

$$S = I - E$$

where:

- S = monthly savings,
- I = monthly income,
- E = monthly expenses.

Savings Rate

$$SR = \frac{I - E}{I} \times 100$$

where SR represents the percentage of income remaining after expenses.

Expense-to-Income Ratio

$$ER = \frac{E}{I} \times 100$$

These indicators provide the foundation for understanding the user's financial capacity.

3.4 Risk Assessment

The system uses a structured questionnaire to assess the user's risk preferences.

The questionnaire can contain approximately 8–10 questions related to:

- reaction to investment losses;
- investment experience;
- investment horizon;
- financial goals;
- liquidity requirements;
- willingness to accept volatility;
- expected investment growth;
- response to market fluctuations.

Each response is assigned a predefined score.

The total risk score can be represented as:

$$R = \sum_{i=1}^n q_i$$

where:

- R = total risk score;
- q_i = score assigned to question i ;
- n = total number of questions.

The resulting score is mapped to a predefined risk category.

Table 3. Risk Classification Framework

Risk Score Range	Risk Profile	General Characteristics
[TO BE DEFINED]	Conservative	Lower tolerance for uncertainty
[TO BE DEFINED]	Moderate	Balanced tolerance for uncertainty
[TO BE DEFINED]	Aggressive	Higher tolerance for uncertainty

Note: The score boundaries will be finalized and justified based on the selected risk-assessment methodology before the final paper submission.

3.5 Investment Category Matching

The recommendation engine uses three principal factors:

Risk Profile + Investment Horizon + Financial Goal

The system does not select an individual stock or guarantee a particular financial outcome. Instead, it identifies a broad investment category.

Table 4. Investment Category Mapping

Risk Profile	Short/Medium Horizon	Long Horizon
Conservative	Lower-risk categories	Lower-risk/diversified categories
Moderate	Balanced categories	Balanced/diversified categories
Aggressive	[Defined according to methodology]	Equity-oriented/diversified growth categories

The final recommendation rules will be defined before evaluation and applied consistently across all test cases.

4. AI AGENT DESIGN

The AI component is implemented as a **single financial decision-support agent** with controlled access to predefined tools.

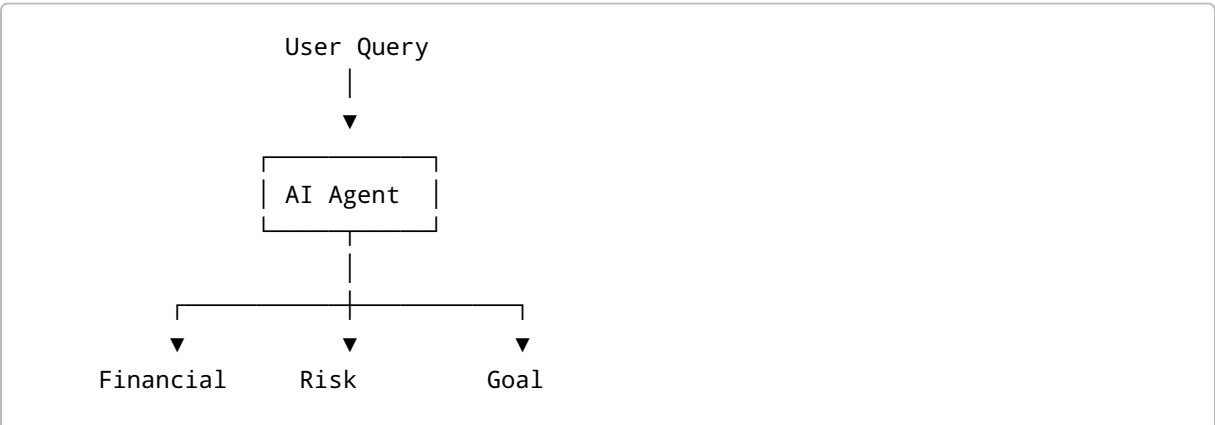
Rather than allowing the language model to independently perform financial calculations, the agent interacts with deterministic tools that return structured outputs.

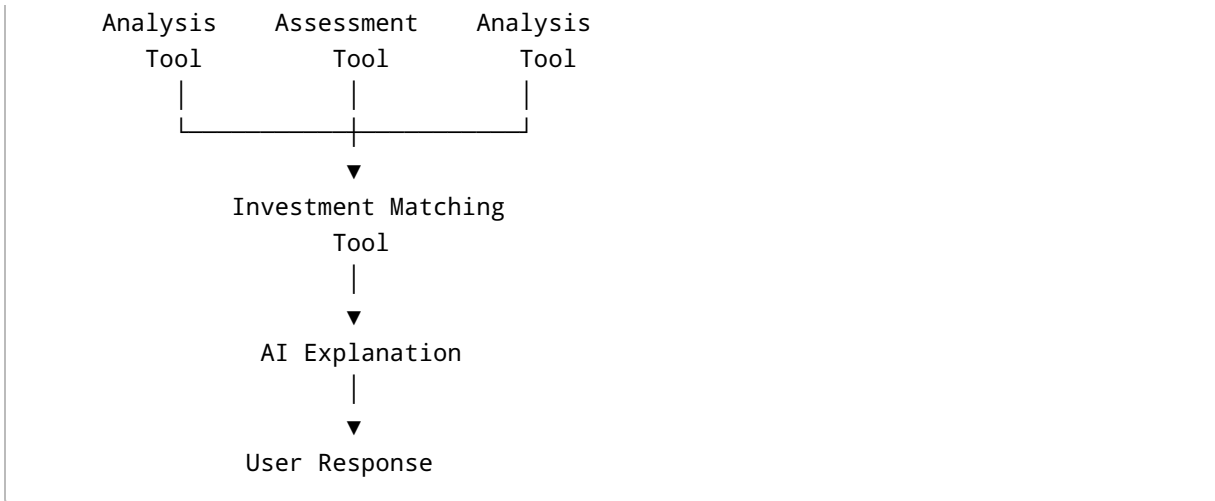
4.1 AI Agent Workflow

Figure 2. AI Agent Tool-Calling Architecture

[INSERT FIGURE 2 HERE]

Figure placeholder:





4.2 Agent Tools

The proposed agent contains five tools.

Tool 1 — Financial Analysis Tool

Calculates:

- savings;
- savings rate;
- expense ratio;
- disposable income;
- investment capacity indicators.

Tool 2 — Risk Assessment Tool

Processes the risk questionnaire and returns:

- risk score;
- risk category;
- relevant assessment factors.

Tool 3 — Goal Analysis Tool

Analyses:

- financial goal;
- investment horizon;
- target amount where applicable;
- required financial capacity.

Tool 4 — Investment Matching Tool

Uses the structured profile to identify a suitable investment category.

Tool 5 — Explanation Tool

Converts the structured analytical output into an understandable explanation.

The agent therefore acts as an **orchestration and explanation layer** rather than replacing the underlying financial calculation mechanisms.

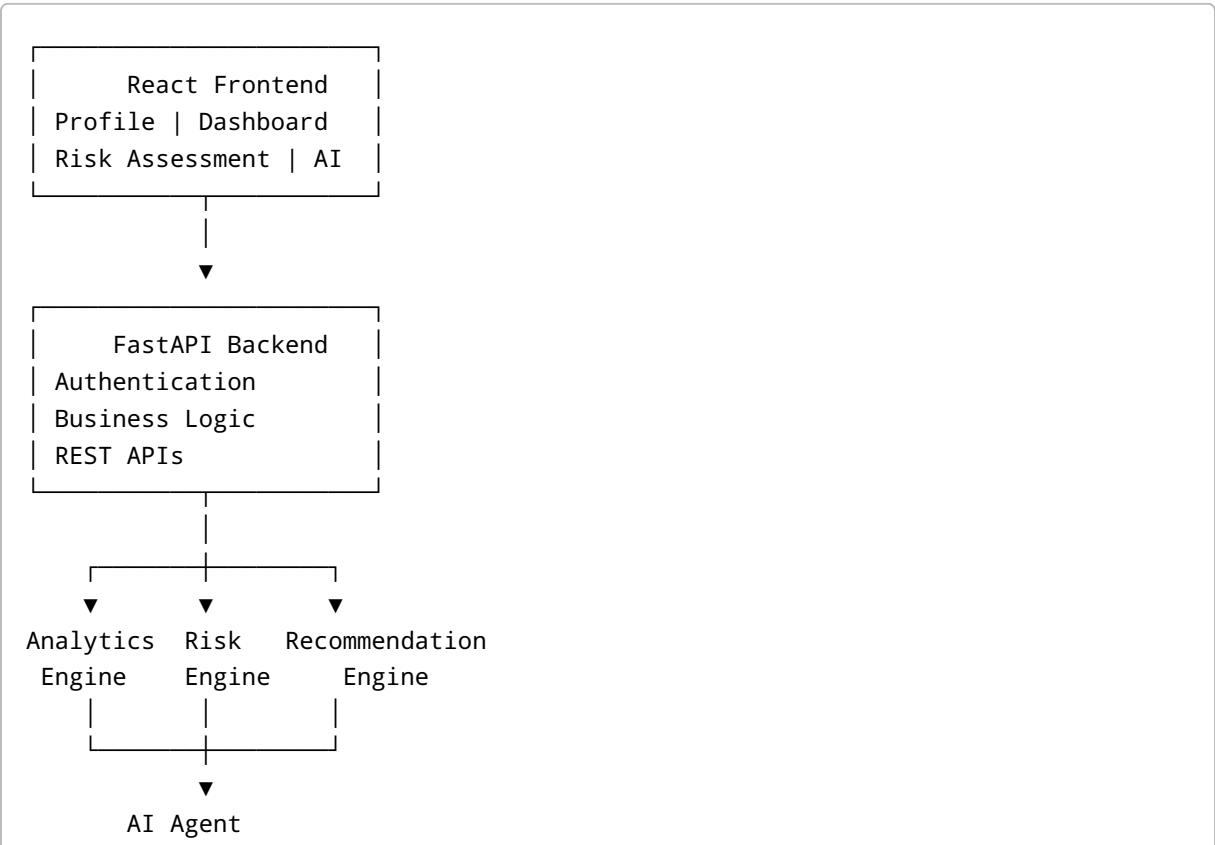
5. SYSTEM ARCHITECTURE

The system follows a client-server architecture consisting of a frontend, backend, database, analytics layer, recommendation engine, and AI agent.

Figure 3. Overall System Architecture

[INSERT FIGURE 3 HERE]

Figure placeholder:





6. TECHNOLOGY STACK

Table 5. Technology Stack

Layer	Technology	Purpose
Frontend	React.js	User interface and dashboard
Backend	Python	Core application logic
API Framework	FastAPI	REST API development
Database	PostgreSQL	User and analytical data
Data Processing	Pandas, NumPy	Data processing and calculations
AI	LLM API	Agent reasoning and explanation
Agent Mechanism	Function/Tool Calling	Controlled tool execution
Visualization	Recharts / Chart.js	Financial charts
API Testing	Postman	Backend API testing
Development	VS Code	Application development
Version Control	Git/GitHub	Source-code management

The original project specification also identifies a modern responsive frontend, backend APIs, database storage, AI integration, function calling, conversation history, and financial analytics as technical requirements.

7. SYSTEM IMPLEMENTATION

7.1 User Profile Module

The user interface provides a form through which users enter the required financial information.

The form includes:

- age;
- monthly income;

- monthly expenses;
- current savings;
- intended investment amount;
- investment horizon;
- financial goal.

Input validation is applied to ensure that negative or invalid financial values are not accepted.

7.2 Financial Analytics Module

The backend receives the user's financial information through a REST API. The analytics module calculates the predefined financial indicators.

The output is returned to the frontend in structured form.

Example Output

Indicator	Value
Monthly Income	₹XX,XXX
Monthly Expenses	₹XX,XXX
Monthly Savings	₹XX,XXX
Savings Rate	XX%
Expense Ratio	XX%

[INSERT DASHBOARD SCREENSHOT HERE]

Figure 4. Financial Analytics Dashboard

7.3 Risk Assessment Module

The risk questionnaire is presented through an interactive interface. Each response is converted into a numerical score.

After submission, the backend:

1. validates responses;
2. calculates the total score;
3. determines the risk category;
4. stores the assessment;
5. passes the result to the recommendation layer.

7.4 Investment Matching Module

The recommendation engine receives the user's:

- risk profile;
- investment horizon;
- financial goal;
- financial capacity.

It then applies predefined rules to identify an appropriate investment category.

The use of explicit rules provides consistency and makes the recommendation mechanism easier to evaluate.

7.5 AI Agent Module

The AI agent receives the user's query and available structured financial information.

Based on the request, the agent determines which tool or tools are required.

For example:

```
User Query
  ↓
"How suitable is this investment approach for me?"
  ↓
AI Agent
  ↓
Risk Assessment Tool
  ↓
Financial Analysis Tool
  ↓
Goal Analysis Tool
  ↓
Investment Matching Tool
  ↓
AI-generated explanation
```

The agent response is generated using the outputs of the tools rather than independently creating numerical financial values.

8. DATABASE DESIGN

The database stores only the information required for the system.

Table 6. Main Database Tables

Table	Important Fields
Users	user_id, age, created_at
Financial_Profile	user_id, income, expenses, savings, investment_amount
Risk_Assessment	assessment_id, user_id, responses, score, profile
Recommendations	recommendation_id, user_id, category, explanation
Conversations	conversation_id, user_id, query, response, timestamp

Figure 5. Database Entity Relationship Diagram

[INSERT ER DIAGRAM HERE]

The database should use appropriate access controls and avoid storing unnecessary sensitive information.

9. EXPERIMENTAL DESIGN

The system will be evaluated using a controlled dataset.

9.1 Dataset

A synthetic dataset containing approximately **[100–500] financial profiles** can be generated for initial testing.

Each record will contain:

- age;
- income;
- expenses;
- savings;
- investment amount;
- investment horizon;
- financial goal;
- risk responses;
- calculated risk score;
- risk category.

Synthetic data allows the system to be tested across different financial conditions without exposing real individuals' financial information.

If real users are included in the evaluation, their information must be collected with appropriate consent and anonymized before analysis.

9.2 Test Cases

Test cases should cover different financial situations.

Table 7. Example Test Profiles

Profile	Income	Expenses	Horizon	Risk Type	Expected Profile
P1	₹40,000	₹35,000	2 years	Low	Conservative
P2	₹70,000	₹45,000	7 years	Medium	Moderate
P3	₹1,00,000	₹55,000	12 years	High	Aggressive
P4	[Value]	[Value]	[Value]	[Value]	[Value]

Note: These are example test cases for development. Final evaluation values should come from the actual dataset.

10. TESTING AND EVALUATION

The proposed system will be evaluated at three levels.

10.1 Functional Testing

Functional testing verifies whether each component operates according to its requirements.

Table 8. Functional Test Cases

Test Case	Expected Result	Status
User registration	Account created successfully	[PASS/FAIL]
Financial data submission	Data validated and stored	[PASS/FAIL]
Savings calculation	Correct value returned	[PASS/FAIL]
Risk assessment	Correct score generated	[PASS/FAIL]
Risk classification	Correct category returned	[PASS/FAIL]

Test Case	Expected Result	Status
Recommendation	Appropriate category returned	[PASS/FAIL]
AI explanation	Explanation generated	[PASS/FAIL]

10.2 Risk Classification Evaluation

The system's risk classification will be compared against the expected classification defined by the selected scoring methodology.

Accuracy can be calculated as:

$$Accuracy = \frac{Correct\ Classifications}{Total\ Test\ Cases} \times 100$$

Table 9. Risk Classification Results

Metric	Result
Total test profiles	[XX]
Correct classifications	[XX]
Incorrect classifications	[XX]
Classification accuracy	[XX%]

10.3 Recommendation Evaluation

The recommendation mechanism can be evaluated by reviewers or domain-informed evaluators using criteria such as:

- risk alignment;
- goal alignment;
- investment-horizon alignment;
- relevance;
- consistency.

A five-point Likert scale can be used.

Table 10. Recommendation Evaluation

Criterion	Mean Score / 5
Risk alignment	[X.XX]

Criterion	Mean Score / 5
Goal alignment	[X.XX]
Horizon alignment	[X.XX]
Relevance	[X.XX]
Consistency	[X.XX]

10.4 AI Explanation Evaluation

The AI component should be evaluated separately from the numerical recommendation.

Possible criteria include:

- clarity;
- relevance;
- completeness;
- factual consistency;
- ease of understanding.

Table 11. AI Explanation Evaluation

Criterion	Mean Score / 5
Clarity	[X.XX]
Relevance	[X.XX]
Understandability	[X.XX]
Completeness	[X.XX]
Factual consistency	[X.XX]

11. RESULTS AND ANALYSIS

This section will be completed after implementation and evaluation. No numerical result should be inserted without experimental evidence.

11.1 Financial Analytics Results

The system will be tested using multiple financial profiles to verify whether the calculated financial indicators match independently calculated reference values.

Table 12. Financial Calculation Validation

Indicator	Reference	System Output	Difference
Monthly savings	[XX]	[XX]	[XX]
Savings rate	[XX%]	[XX%]	[XX%]
Expense ratio	[XX%]	[XX%]	[XX%]

[INSERT GRAPH HERE]

Figure 6. Comparison of Reference and System-Calculated Financial Indicators

11.2 Risk Profiling Results

The risk-profiling evaluation will report the number of correctly and incorrectly classified profiles.

[INSERT GRAPH HERE]

Figure 7. Distribution of Users Across Risk Profiles

Suggested graph:

Conservative | Moderate | Aggressive

11.3 Recommendation Results

The recommendation evaluation will show whether the generated investment categories are aligned with the defined risk, goal, and horizon criteria.

[INSERT GRAPH HERE]

Figure 8. Average Recommendation Evaluation Scores

11.4 AI Agent Results

The AI agent will be evaluated based on whether it correctly uses analytical outputs and provides understandable explanations.

Examples of evaluation questions:

1. Did the agent use the correct financial information?
 2. Did the agent preserve the calculated risk profile?
 3. Did the explanation match the recommendation?
 4. Did the response avoid unsupported numerical claims?
 5. Was the explanation understandable to the user?
-

12. DISCUSSION

The proposed system is designed to demonstrate how financial analytics can serve as the foundation for AI-assisted financial decision support. Rather than allowing an AI model to independently generate investment recommendations, the system separates numerical analysis, risk classification, recommendation logic, and natural-language explanation.

This separation provides two advantages. First, financial calculations remain deterministic and can be independently validated. Second, the AI agent can focus on interpreting and communicating structured analytical outputs.

The use of a tool-based agent also allows the system to respond to different user queries using the appropriate analytical capability. For example, a question related to financial capacity can invoke the financial-analysis tool, while a question concerning risk tolerance can invoke the risk-assessment tool.

The results obtained from the evaluation will determine whether the proposed approach provides consistent risk classification, relevant investment-category matching, and useful AI-generated explanations.

[ADD 1–2 PARAGRAPHS HERE AFTER RESULTS ARE AVAILABLE]

The discussion should compare the findings with the studies identified in the literature review and explain whether the proposed approach addresses the identified research gap.

13. SECURITY, PRIVACY AND RESPONSIBLE AI

Financial information requires careful handling because it may contain sensitive personal and economic information.

The proposed system should therefore incorporate:

- authenticated access;
- secure password storage;
- input validation;
- controlled database access;

- encrypted communication;
- minimal collection of personal information;
- anonymization of research datasets.

The AI component should also operate within clearly defined boundaries. It should not claim guaranteed investment returns or present generated content as a substitute for professional financial advice.

The recommendation mechanism should remain transparent by allowing users to understand the principal factors influencing the result.

The system is therefore intended as a **decision-support application**, not an autonomous financial management system.

14. LIMITATIONS

The proposed study has several limitations.

First, the initial evaluation may rely on synthetic financial profiles. Although synthetic data is useful for controlled testing, it may not fully represent the diversity of real-world financial situations.

Second, risk profiling is partly dependent on self-reported preferences. A user's stated tolerance for risk may differ from actual behaviour during periods of financial uncertainty.

Third, the system provides investment-category decision support rather than predicting future market performance. Therefore, recommendation suitability does not imply future financial returns.

Fourth, AI-generated explanations may contain language-generation errors. The use of deterministic analytical tools reduces this risk for numerical outputs, but the natural-language layer still requires appropriate validation.

Finally, the evaluation period may not be sufficient to determine long-term changes in user investment behaviour or financial outcomes.

15. FUTURE SCOPE

The proposed system can be extended in several directions.

Future work may include:

1. Integration with larger real-world financial datasets.
2. Expert validation of the risk-profiling methodology.
3. More advanced personalization techniques.

4. Real-time financial-market information.
5. Portfolio monitoring.
6. Mutual-fund analysis.
7. Advanced machine-learning-based recommendation models.
8. Longitudinal user studies.
9. Explainable AI techniques for financial recommendations.
10. Mobile application development.

These extensions can improve the capabilities of the system while retaining the core financial analytics and AI-agent architecture.

16. CONCLUSION

This paper presents an AI-assisted approach for financial risk profiling and personalized investment decision support. The proposed system combines structured financial analytics, risk assessment, investment-category matching, and a tool-using AI agent to transform personal financial information into understandable decision-support insights.

The financial analytics component calculates indicators such as savings and expense ratios, while the risk-assessment component determines an individual's risk profile using structured questionnaire responses. A controlled recommendation mechanism then maps the user's financial characteristics, investment horizon, and risk profile to an appropriate investment category. The AI agent operates as an orchestration and explanation layer, using predefined analytical tools and communicating the resulting insights in natural language.

The proposed architecture deliberately separates deterministic financial calculations from AI-generated explanations. This provides a more controlled approach than allowing a language model to independently perform financial calculations or generate unrestricted investment recommendations.

The final evaluation will determine the effectiveness of the system in terms of risk-classification consistency, recommendation relevance, explanation quality, and usability. Overall, the project demonstrates how financial analytics and AI agents can be combined to provide personalized and interpretable financial decision support while maintaining appropriate boundaries around automated financial decision-making.

REFERENCES

[References to be populated after the literature review using verified academic sources and formatted according to APA 7th edition.]

Suggested literature categories:

1. Financial analytics and personal finance
2. Investment risk profiling

3. AI-based financial decision support
 4. Robo-advisory systems
 5. AI agents and tool/function calling
 6. Explainable AI
 7. Personalized recommendation systems
 8. Responsible AI in financial services
-

APPENDIX — OPTIONAL

Appendix A. Risk Assessment Questionnaire

Q1. How would you react if your investment temporarily lost 20% of its value?

1. Withdraw immediately
2. Become uncomfortable and consider withdrawing
3. Wait until the situation improves
4. Continue with the investment
5. Consider investing more

Q2. What is your expected investment horizon?

1. Less than 1 year
2. 1–3 years
3. 3–5 years
4. 5–10 years
5. More than 10 years

Q3. What is your primary financial objective?

- Capital preservation
- Income generation
- Balanced growth
- Long-term wealth creation

Q4–Q10. Additional questions addressing investment experience, liquidity requirements, volatility tolerance, financial goals, and investment expectations.

Recommended Figures for the 7–8 Page Paper

To keep the paper within the page limit, I would use **5–6 figures maximum**:

Figure	Content
Figure 1	Overall proposed methodology

Figure	Content
Figure 2	AI agent + five tools
Figure 3	System architecture
Figure 4	Financial analytics dashboard
Figure 5	Risk assessment/recommendation screen
Figure 6	Results graph

And approximately **8-10 tables**:

1. Existing vs proposed approach
2. Input variables
3. Risk classification
4. Investment category mapping
5. Technology stack
6. Database design
7. Test profiles
8. Functional testing
9. Risk classification results
10. Recommendation/AI evaluation

This gives us a **real 7-8 page conference paper**, rather than a long project report. The original project's broader features can be treated as future extensions; the current research remains tightly focused on **financial analytics + one AI agent + risk profiling + investment decision support**.